
Response to Reviewer cpjj:

Thank you very much for your valuable feedback.

Weakness 1:

Knowledge distillation with the ViT architecture is considered a more challenging task. Several studies have been conducted by researchers on this issue, including Cross-Architecture Knowledge Distillation, Learning Efficient Vision Transformers via Fine-Grained Manifold Distillation, ViTKD [1], and One-for-All [2], etc.

References:

[1] Yang, Z., Li, Z., Zeng, A., Li, Z., Yuan, C., & Li, Y. (2022). Vitkd: Practical guidelines for vit feature knowledge distillation. arXiv preprint arXiv:2209.02432.

[2]: Hao, Z., Guo, J., Han, K., Tang, Y., Hu, H., Wang, Y., & Xu, C. (2024). One-for-All: Bridge the Gap Between Heterogeneous Architectures in Knowledge Distillation. Advances in Neural Information Processing Systems, 36.

Weakness 2:

Our proposed method has achieved state-of-the-art results on large-scale MS-COCO datasets (COCO Stuff164k) and ADE 20k, as indicated in the Table A in **Official Comment**.

Weakness 3:

We present visual results on the Cityscapes, Pascal VOC, Synapses, and FLARE22 datasets. We have highlighted the challenging regions in Figure 5-8, such as pillars in Cityscapes, bird feet in Pascal VOC, pancreas in the Synapses and the duodenum the FLARE22.

Weakness 4:

The limitations of our method include: (1) In Table 3, there is no significant improvement in the mHD95 metric (related to boundary) when the teacher is TransUNet and the student is FCN-ERFNet; (2) The hierarchical region fusion ability of HRDKD may require further improvement, especially on datasets like Pascal VOC; (3) Our follow-up research is needed on cross-architecture knowledge transfer in natural image semantic segmentation, specifically between ViT Transformer and CNN architectures.

Weakness 5:

The reason why our proposed HRDKD performs worse than RDKD on the Pascal VOC dataset may be due to the presence of a large number of background pixels in the Pascal VOC labels, which leads to unfavorable effects during aggregation. However, for the fully pixel-level segmentation datasets, Cityscapes, Synapses, and FLARE22, HRDKD generally outperforms RDKD.

Question 1:

We have already explained this in Weakness 4.

Question 2:

We have already explained this in Weakness 5.

Question 3:

The loss weight parameters for Formula 6 are introduced in line 281 of the paper, while the loss weight parameters for Formula 7 are described in line 296.

Question 4:

No, because the proposed HRDKD can be considered an extension of the RDKD method, and HRDKD involves multiple aggregations of model predictions, each associated with a calculation of RDKD loss.

Question 5:

The term *dark knowledge* is mentioned in the DKD literature, referring to the knowledge transferred from a teacher model to a student model. In DKD, this is expressed as follows: *The dark knowledge is transferred to students via soft labels from teachers.*